SECTION-I

Single Correct Answer Type **7 Q. [3 M (–1)]**

1. Current I flowing along the path ABCD, along the four edges of the cube (figure-a), creates a magnetic field in the centre of the cube of B_0. Find the magnetic field B created at the center of the cube by a current I flowing along the path of the six edges ABCGHEA (figure-b)

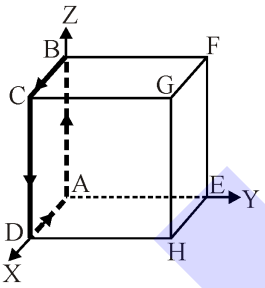


Figure-(a)

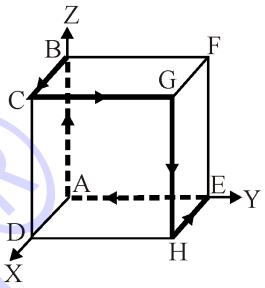


Figure-(b)

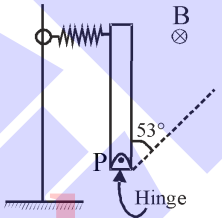
(A) $\sqrt{\frac{3}{2}}B_0$ Towards corner G

(B) $\sqrt{3}B_0$ Towards corner E

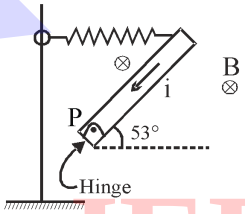
(C) $\sqrt{\frac{3}{2}}B_0$ Towards corner H

(D) $\sqrt{3}B_0$ Towards corner F

2. A thin uniform rod with negligible mass and length ℓ is attached to the floor by a frictionless hinge at point P. A horizontal spring with force constant k connects the other end to wall. The rod is in a uniform magnetic field B directed into the plane of paper. What is extension in spring in equilibrium when a current *i* is passed through the rod in direction shown. Assuming spring to be in natural length initially.



Spring in relaxed position



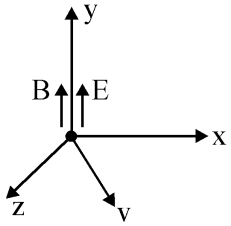
In equilibrium position

(A) $\frac{5i\ell B}{8k}$ (B) $\frac{3i\ell B}{8k}$ (C) $\frac{5i\ell B}{4k}$ (D) $\frac{5i\ell B}{6k}$

3. A positive charge particle having charge q and mass m has velocity $\vec{v} = v\left(\frac{\hat{i}+\hat{k}}{\sqrt{2}}\right)$ in the magnetic field at the origin. Its speed as the function of y is :-

(A) $\sqrt{v^2 + \frac{qE}{2m}y}$

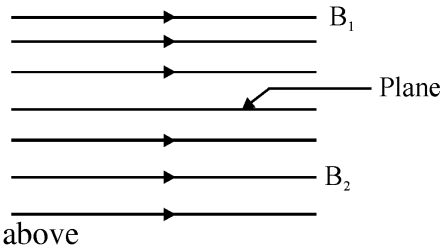
(B) $\sqrt{\left(\frac{B}{E}\right)^2 + v^2 + \frac{qE}{2m}y}$



(C) $\sqrt{v^2 + \frac{2qE}{m}y}$

(D) None of the above

4. Two uniform magnetic fields B_1 and B_2 exist above and below the plane. The magnetic force acting on unit area of the plane is :-



(A) $\frac{B_2^2 - B_1^2}{2\mu_0}\uparrow$

(B) $\frac{|B_1 - B_2|^2}{2\mu_0}\downarrow$

(C) $\frac{\left|\frac{B_1 + B_2}{2}\right|^2}{2\mu_0}\uparrow$

(D) None of the above

5. To increase the current sensitivity of a moving coil galvanometer, we should :–

(A) decrease number of turns in coil (B) decrease area of cross section of the coil

(C) increase torsional constant of spiral springs (D) None of the above

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6. A straight conductor of mass 100gm and length 20cm is placed on the xy plane in a gravity free space. A uniform and constant magnetic field of 1000T in positive z direction is applied on it. A charge of 1mC is passed through the rod in a very short time duration and the conductor begins to move just after all the charge has passed through it. Assuming no friction, the velocity of the conductor is :

(A) 2cm/s (B) 2 m/s (C) 0 (D) 20 cm/s

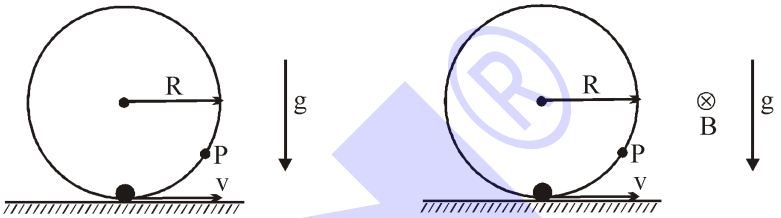
7. In case figure (a) and (b) a positive charge particle of mass m and charge q is given velocity 'v' at the lowest point of a smooth fixed sphere of radius R. In both cases circular motion is in the plane of the paper. In case (b) magnetic field is also switched on in inward direction. If at point P, v_1 and v_2, N_1 and N_2, are velocities and normal forces in two cases respectively then.

(A) $v_1 = v_2$ and $N_1 > N_2$

(B) $v_1 < v_2$ and $N_1 > N_2$

(C) $v_1 > v_2$ and $N_1 > N_2$

(D) $v_1 = v_2$ and $N_1 = N_2$



(a) $v = \sqrt{2gR}$ (b) $v = \sqrt{2gR}$

Multiple Correct Answer Type 9 Q. [4 M (–1)]

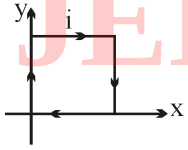
8. Figure shows a square current carrying coil of edge length L. The magnetic field on the coil is given by $\vec{B} = \frac{B_0 y}{L}\hat{i} + \frac{B_0 x}{L}\hat{j}$ where B_0 is a positive constant.

(A) If coil is free to rotate about x axis torque on the coil is given by $\frac{1}{2}iAB_0\hat{i}$.

(B) If coil is free to rotate about y - axis torque on coil is given by $-\frac{1}{2}iAB_0\hat{j}$.

(C) Resultant force on coil is zero.

(D) Equation for the torque $\vec{\mu} \times \vec{B}$ where μ is magnetic moment of coil is not valid on the coil.



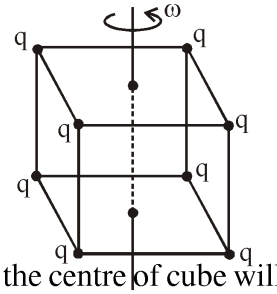
9. Consider a cube of side 'a' as shown. Eight point charges are placed at the corners. The cube is rotated about the axis with constant angular velocity 'ω':

(A) Net magnetic field at the centre of cube is zero

(B) Net magnetic field at the centre of cube is $\frac{\sqrt{2}\mu_0 q\omega}{\pi a}$

(C) Net magnetic field at the centre of cube is $\frac{8}{3\sqrt{3}}\frac{\mu_0 q\omega}{\pi a}$

(D) If polarity of any four charges are reversed, then magnetic field at the centre of cube will be zero.



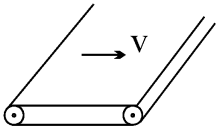
10. Charge is sprayed onto a large non conducting belt above the left hand roller. The belt carries charge with a uniform surface charge density σ, as it moves with a speed v between the rollers as shown. The charge is removed by a wiper at right hand roller. For a point just above the sheet mark the correct option.

(A) magnetic field is $\frac{\mu_0 \sigma v}{2}$, out of the plane of the page, parallel to axis of roller.

(B) magnetic field is $\mu_0\sigma$, out of the plane of the page, perpendicular to axis

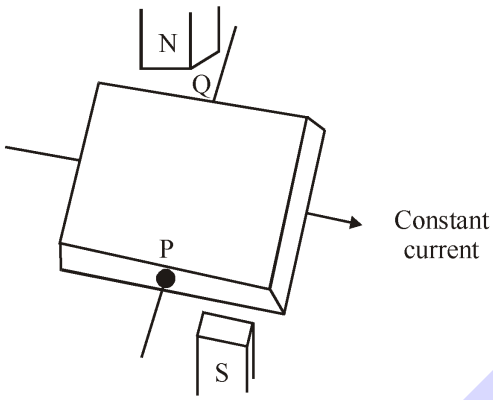
(C) electic field is $\frac{\varepsilon_0 \sigma}{2}$ perpendicular to the plane of sheet

(D) If an electron moves parallel to V just above the sheet it will experience an upward magnetic force.



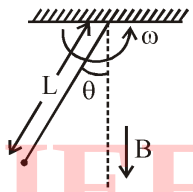
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11. Figure shows the essential parts of an apparatus to demonstrate the Hall effect. Which of the following statement (s) is/are **CORRECT** ?



(A) In the arrangement above, the Hall potential difference is developed across PQ
(B) The magnitude of the Hall potential is greater if the applied magnetic flux density is increased
(C) The magnitude of the Hall potential is less if the width PQ of the specimen is decreased
(D) Hall potential is independent of material used for conductor

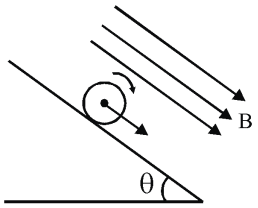
12. A charged particle having its charge to mass ratio as β goes in a conical pendulum of length L making an angle θ with vertical and angular velocity ω. If a magnetic field B is directed vertically downwards (see figure) :



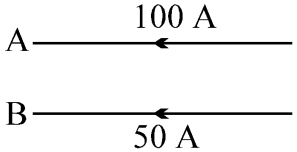
(A) $B = \frac{1}{\beta}\left[\omega - \frac{g}{\omega L\cos\theta}\right]$
(B) Angular momentum of the particle about the point of suspension remains constant.
(C) If the direction of B were reversed maintaining same ω and L, then θ will remain unchanged.
(D) Rate of change of angular momentum of the particle about the point of suspension is not a constant vector.

13. A cylindrical wire is rolling on rough inclined plane. There exist a magnetic field along the plane as shown. At a certain instant, when its angular velocity is ω, a current i is switched on across the length of cylinder in such a way that its angular velocity becomes constant. Find the **CORRECT** statement(s) :-

(A) translational velocity will also become constant
(B) normal reaction will become zero
(C) friction force will become zero
(D) force on the cylinder due to magnetic field will be mgcosθ.



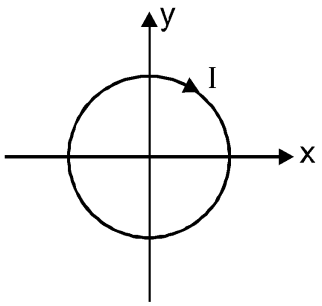
14. Two long parallel conductors are carrying currents in the same direction as shown in the figure. The upper conductor (A) carries a current of 100 A and is held firmly in position. The lower conductor (B) carries a current of 50 A and is free to slide up and down. The linear mass density of the lower conductor is 0.01 kg m^{-1}.



(A) Conductor B will be in equilibrium if the distance between the conductors is 0.01 m^{-1}.
(B) Conductor B performs SHM with period $\frac{\pi}{50}$ s.
(C) Equilibrium of conductor B is stable.
(D) Equilibrium of conductor B is unstable.

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15. A uniform circular loop of mass 'm' and radius R is lying in space where a uniform magnetic field $\vec{B} = B_0(\hat{i} + \sqrt{3}\,\hat{j})$. The loop carries constant current I in clockwise direction and lies in a gravity free region at rest in xy–plane with its centre at origin. Then choose the CORRECT statement(s) :-



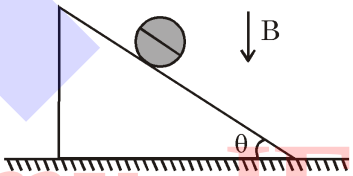
(A) The initial angular acceleration of the loop has magnitude $\frac{4I\pi B_0}{m}$.

(B) The initial angular acceleration of the loop has magnitude $\frac{2I\pi B_0}{m}$

(C) The unit vector in the direction of initial angular acceleration is $\left(+\frac{\sqrt{3}}{2}\hat{i} - \frac{1}{2}\hat{j}\right)$

(D) The initial acceleration of the point with coordinate $\left(\frac{\sqrt{3}}{2}R, \frac{-R}{2}\right)$ is zero

16. In the figure shown a coil of single turn is wound on a sphere of radius R and mass M. The plane of the coil is parallel to the inclined plane and lies in the equatorial plane of the sphere. Current in the coil is i. The sphere is in equilibrium.



(A) The value of B is $\frac{mg\tan\theta}{\pi iR}$

(B) The current in coil is clockwise as seen from top.

(C) Minimum friction coefficient required is $\tan\theta$.

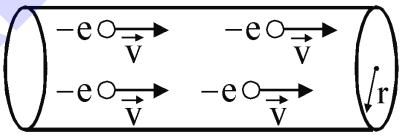
(D) If the surface is smooth, sphere will slide down without rotation.

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Linked Comprehension Type (1 Para × 3Q.) (1 Para × 2Q.) [3 M (-1)]
(Single Correct Answer Type)

Paragraph for Questions no. 17 to 19

A beam of electrons has radius r and contains n electrons per cubic meter moving with velocity $\vec{v}$ along the beam (figure). Assume that a beam is much longer than its diameter forms a cylindrically symmetric distribution of charge and current. The beam expand if the electric force exceeds the magnetic force. So long as the expansion is slow, the approximation of cylindrical symmetry and static fields remain valid.



17. What is electric field produced on the edge of the beam?

(A) $\vec{E} = \frac{ner}{4\varepsilon_0}\hat{r}$ (B) $\vec{E} = \frac{ner}{2\varepsilon_0}\hat{r}$ (C) $\vec{E} = \frac{-ner}{4\varepsilon_0}\hat{r}$ (D) $\vec{E} = \frac{-ner}{2\varepsilon_0}\hat{r}$

18. What is the magnetic field produced on the edge of the beam?

(A) $\frac{ner}{4}\mu_0 v$ (B) $\frac{ner}{2}\mu_0 v$ (C) $ner\mu_0 v$ (D) $\frac{ner}{6}\mu_0 v$

19. What is net force on the electron at the edge of the beam?

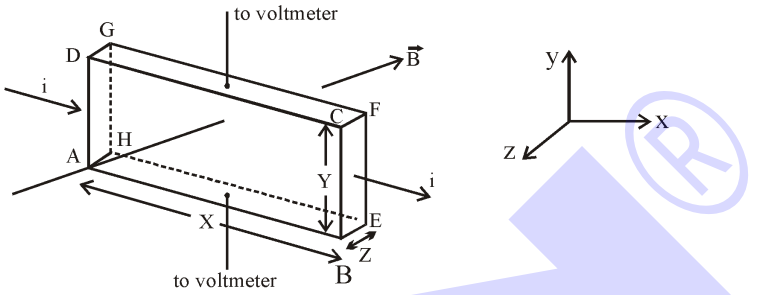
(A) $\frac{2ne^2r}{\varepsilon_0}\left(1-\mu_0\varepsilon_0 v^2\right)\hat{r}$ (B) $\frac{ne^2r}{2\varepsilon_0}\left(1-\mu_0\varepsilon_0 v^2\right)\hat{r}$ (C) $\frac{4ne^2r}{\varepsilon_0}\left(1-\mu_0\varepsilon_0 v^2\right)\hat{r}$ (D) $\frac{ne^2r}{4\varepsilon_0}\left(1-\mu_0\varepsilon_0 v^2\right)\hat{r}$

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Paragraph for Question no. 20 and 21

When a conductor carries a current i the charge carriers move with a drift velocity v_d directed along the length of conductor. A uniform magnetic field B is applied perpendicular to current as shown in figure.

Magnetic force deflects charge carriers. As a resultant of magnetic deflection a potential difference called Hall voltage is established in transverse direction to current. In usual notations n = number of electron per unit volume, σ = conductivity of conductor, q = magnitude of charge carrier.



20. Mark the **CORRECT** option :

(A) In steady state face CFGD becomes positively charged

(B) Hall voltage developed across conductor has magnitude $\frac{1}{nq}\frac{iB}{Y}$

(C) An electric field is established from face ABCD towards EFGH

(D) By decreasing Z, voltage developed can be increased.

21. Direction of resultant electric field inside conductor from x-axis is given by :

(A) $\tan\alpha = \frac{\sigma B}{nq}$ (B) $\tan\alpha = \frac{iB}{\sigma n}$

(C) $\tan\alpha = \frac{\sigma n}{iB}$ (D) $\tan\alpha = \frac{ni}{\sigma B}$

SECTION-II

Numerical Answer Type Question **2Q.[3M (0)]**

(upto second decimal place)

22. Two coaxial long solenoids having number of turns 3000 & 1500 per metre, radii 2.0 m & 1.0 m and carry current of 2A & 4A in opposite sense respectively. Find the magnetic energy stored per unit length (in J/m). [Take $\pi^2 = 10$]

23. A 1.5 m diameter cyclotron is to accelerate protons to an energy of 8 MeV. The magnetic field strength for this cyclotron required is given by B tesla. Fill 300 B in OMR sheet. (Given mass of proton = 1.6×10^{-27} kg)

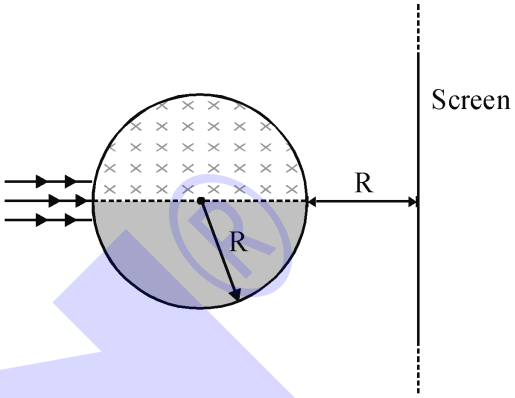
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SECTION-III

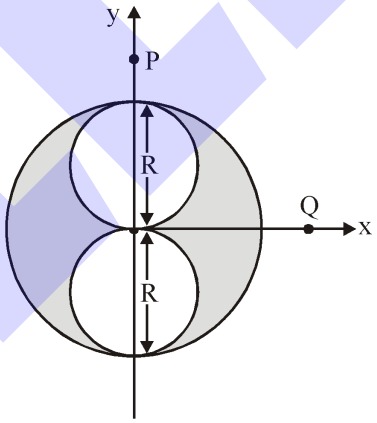
Numerical Grid Type (Ranging from 0 to 9) **4 Q. [4 M (0)]**

24. Figure shows circular region of radius R = $\sqrt{3}$ m in which upper half has uniform magnetic field $\vec{B} = 0.2(-\hat{k})T$ and lower half has uniform magnetic field $\vec{B} = 0.2\hat{k}T$. A very thin parallel beam of point charges each having mass m = 2 gm, speed v = 0.3 m/sec and charge q = +1 mC are projected along the diameter as shown in figure. A screen is placed perpendicular to initial velocity of charges as shown. If the distance between the points on screen where charges will strikes after being deflected by the magnetic field is 4X meters, then calculate X.



25. A long, cylindrical conductor of radius R has two cylindrical cavities of diameter R running parallel to its axis as shown in figure. The conductor carries a uniform current density J. The field strength B at position Q (x = 2R) along the x-axis is $\frac{\alpha}{68}\mu_0 JR$ then find α.

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26. The current density inside a long solid cylindrical wire of radius a is in the direction of the central axis and varies linearly with radial distance r from the axis according to $J = J_0\frac{r}{a}$. If the magnetic field inside wire is given by $\frac{\mu_0 J_0 r^x}{ya}$ then find value of x + y.

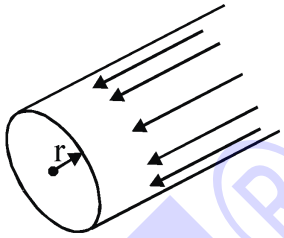
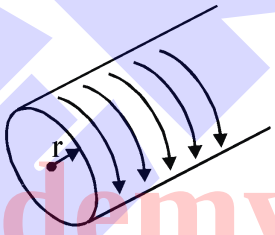
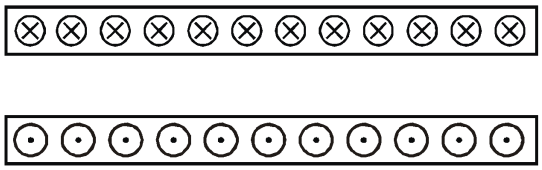
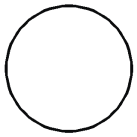
27. An infinite hollow cylinder of radius R having uniform surface charge density σ is rotated with constant angular velocity ω about about its axis. Magnetic field at the centre of cylinder is $B = \mu_0\sigma R^x\omega$. Find x

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SECTION-IV

Matrix Match Type (4 × 5) **1 Q. [8 M (for each entry +2(0)]**

28. Match the column.

	Column-I		**Column-II**
(A)	The pressure on walls due to quantity 'X' is tending to contract them	(P)	Current (X) is flowing axially on the surface of a long thin hollow pipe of radius r.  Energy : Magnetic field energy
(B)	The pressure on walls due to quantity 'X' is tending to expand them	(Q)	Current (X) is flowing tangentially on the surface of a long thin hollow pipe of radius r.  Energy : Magnetic field energy
(C)	The energy/volume inside the walls is more than outside.	(R)	Two current (X) carrying infinite thin sheets as shown in figure.  Energy : Magnetic field energy.
(D)	The energy /volume outside the walls is more than inside.	(S)	A soap bubble present in air containing air. The temperature inside and outside air is the same. Surface tension is (X)  Energy : internal energy of air
		(T)	Point source of light (X) is present at the centre of a spherical shell with both surfaces perfectly black. Energy : light energy.

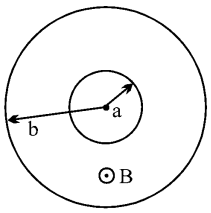
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Subjective Type

2 Q. [4 M (0)]

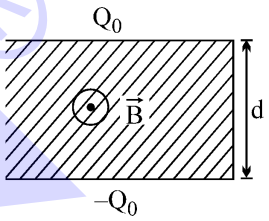
29. Outer cylinder of the coaxial conductor of radius 'b' is given a positive potential V relative to the inner cylinder of radius 'a' as shown in the fig. A static magnetic field B parallel to the axis of the cylinder, directed out of the plane of the paper is also present.



(i) An electron (mass = m_e) is set free with negligible velocity at the surface of the inner cylinder under condition $V = V_0$ and $B = 0$. Find the velocity, when it hits the outer cylinder.

(ii) An electron (mass = m_e) starts out with an initial velocity u_0 in the radial direction under condition $V = 0$ and $B \neq 0$. Find the critical field B_C for which electron will not reach the outer cylinder.

30. A slab is inserted between two parallel plates of a capacitor charged to Q_0. The capacitor is discharged through the slab. A magnetic field is present throughout and is out of the plane of the paper. The total momentum given to the slab after complete discharge is_____.



GR # MAGNETIC EFFECT OF CURRENT — **ANSWER KEY**

SECTION-I

Single Correct Answer Type — **7 Q. [3 M (–1)]**

1. Ans. (D) **2. Ans. (A)** **3. Ans. (C)** **4. Ans. (A)** **5. Ans. (D)**
6. Ans. (B) **7. Ans. (A)**

Multiple Correct Answer Type — **9 Q. [4 M (–1)]**

8. Ans. (ABCD) **9. Ans. (C,D)** **10. Ans. (A, D)** **11. Ans. (A, B)** **12. Ans. (A,D)**
13. Ans. (B,C,D) **14. Ans. (A,D)** **15. Ans. (A, C, D)** **16. Ans. (B,C)**

Linked Comprehension Type (1 Para × 3Q.) (1 Para × 2Q.) [3 M (-1)]
(Single Correct Answer Type)

17. Ans. (D) **18. Ans. (B)** **19. Ans. (B)** **20. Ans. (D)** **21. Ans. (A)**

SECTION-II

Numerical Answer Type Question (upto second decimal place) — **2Q.[3M (0)]**

22. Ans. 216.00 **23. Ans. 160.00**

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) — **4 Q. [4 M (0)]**

24. Ans. 3 **25. Ans. 9** **26. Ans. 5** **27. Ans. 1**

SECTION-IV

Matrix Match Type (4 × 5) — **1 Q. [8 M (for each entry +2(0)]**

28. Ans. (A) P,S; (B) Q,T,R; (C) Q,T,R; (D) P

Subjective Type — **2 Q. [4 M (0)]**

29. Ans. (i) $u = \sqrt{\frac{2eV_0}{m_e}}$, (ii) $\frac{2b\,m_e\,u_0}{e(b^2-a^2)}$

30. Ans. dBQ_0

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